Yuxuan Yang

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Education

Southwestern University of Finance and Economics (SWUFE) **Chengdu, Sichuan, China**
Bachelors of Engineering in Computer Science *August 2018-June 2022*

Washington University in Saint Louis (WUSTL) **Saint Louis, MO**
Master of Science in Computer Science *August 2023-June 2025*

Washington University in Saint Louis (WUSTL) **Saint Louis, MO**
PhD in Computer Science *August 2025-Present*

Work Experience

Southwestern University of Finance and Economics **Chengdu, Sichuan, China**
Research Assistant for SWUFE MLDE Group *August 2022-June 2023*

- Conducted research under professors' guidance.
- Developed a code hub for continual learning baselines with over 2000 lines of code.
- Trained over ten first-year master and undergraduate students in fundamental deep learning.

Research Projects

Learning Control Barrier Functions from Expert Demonstrations using Inverse Constraint Learning
Researcher with Prof. Hussein Sibai *January 2025-September 2025*

- Proposed a data-driven framework that learns neural Control Barrier Functions (CBFs) from expert demonstrations when the failure set is hard to specify formally. Used Inverse Constraint Learning (ICL) to infer a constraint classifier from demonstrations, then leveraged it to label simulated trajectories for neural CBF training. Evaluated on four environments including autonomous driving scenarios, outperforming existing baselines and matching ground-truth-labeled CBF performance.

Designing Latent Safety Filters using Pre-Trained Vision Models
Researcher with Prof. Hussein Sibai *April 2025-September 2025*

- Investigated the use of pre-trained vision models (PVRs) as perception backbones for vision-based safety filters. Applied PVRs to failure set classifiers and Hamilton-Jacobi reachability-based safety filters with latent world models. Analyzed trade-offs between training from scratch, fine-tuning, and freezing PVR weights across robotics control settings.

Forgetting Prevention for Cross-regional Fraud Detection with Heterogeneous Trade Graph
Researcher with Prof. Xin Yang *December 2021-March 2022*

- Proposed a novel solution based on heterogeneous trade graphs, namely Heterogeneous Trade Graph learning for Cross-regional Fraud Detection (HTG-CFD), to prevent knowledge forgetting of cross-regional fraud detection and continuously detect fraudulent transactions in the process of business expansion in new regions.

Awards

WUSTL Dean's Select PhD Fellowship *October 2024*
SWUFE University-level Outstanding Graduation Thesis *June 2022*
National Top 20 in the 2020 Citibank Financial Innovation Application Competition *May 2021*
Honorable Mention in the 2021 Interdisciplinary Contest In Modeling *April 2021*

Skills & Interests

Programming: Python, Mathematica, Java, C/C++
Technologies/Frameworks: Pytorch, Tensorflow, DGL
Interests: Cooking, Music, Game